

Relativity and Electromagnetism Exercise 2: 2026 Feb 18

Problem 1. Consider the Ampère's law in differential and in integral forms. In the differential form it reads,

$$\nabla \times \mathbf{B} = \mu_0 \mathbf{J}.$$

This is a vector equation, or viewed differently, a set of *three* equations for each of the three vector components. In the integral form the equation is

$$\oint_{\partial S} \mathbf{B} \cdot d\mathbf{l} = \mu_0 I_{\text{enclosed}}.$$

This is however a scalar equation. In class I claimed that the two forms are equivalent. How can a set of three equations be equivalent to a single scalar equation?

Problem 2. Consider a steady current I flowing in the z direction. In class I claimed that the corresponding magnetic field $\mathbf{B}$ only has a $\hat{\phi}$ component in cylindrical coordinates. The ϱ component has to be zero in order to respect the equation $\nabla \cdot \mathbf{B} = 0$. Why is the z component zero? The answer lies in symmetry. But why? The current is flowing in the $\hat{z}$ direction as opposed to the $-\hat{z}$ direction; so claiming reflection symmetry about the x - y plane is not permissible. [*Hint:* Consider a charge particle moving radially outward in this field configuration.]

Problem 3. The Biot-Savart law reads,

$$\mathbf{B}(\mathbf{r}) = \frac{\mu_0}{4\pi} \int \frac{d\mathbf{I} \times \mathbf{r}}{r^3} = \int \frac{\mu_0 I}{4\pi} \frac{d\mathbf{l} \times \hat{\mathbf{r}}}{r^2}.$$

Use the Biot-Savart law to obtain the magnetic field for the current configuration in Problem 2. Note: This establishes that the Biot-Savart law is consistent with the Ampère's law.

Problem 4. For the magnetic field configuration in Problem 2 / Problem 3, calculate the divergence and the curl.

Problem 5. Consider two infinite straight line charges λ , a distance d apart, moving along at a constant speed v . How large would v have to be in order for the magnetic attraction to balance the electrical repulsion? Work out the actual number.

